Shear Bands with Dilatancy modelled with Underworld

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DOI https://doi.org/10.6084/m9.figshare.33193365

Keywords Geodynamics, Underworld Code

In honour of Hans Mühlhaus' 70th birthday this month, here are some shear band simulations made with Underworld. We are investigating the role of dilatancy in the geometry of the shear bands for a box of material when a small trapdoor is opened.

The extent to which large-scale deformation is needed to release material through the trapdoor depends on how much the material dilates when shear bands form. This is a taster:

SOURCE https://www.underworldcode.org/shear-bands-with-dilatancy-modelled-with-underworld/
ARCHIVED 2026-08-10T04:19:43Z

VERSION 1.0.0
ARTICLE UWTN 2017-001
PUBLISHED May 05, 2017

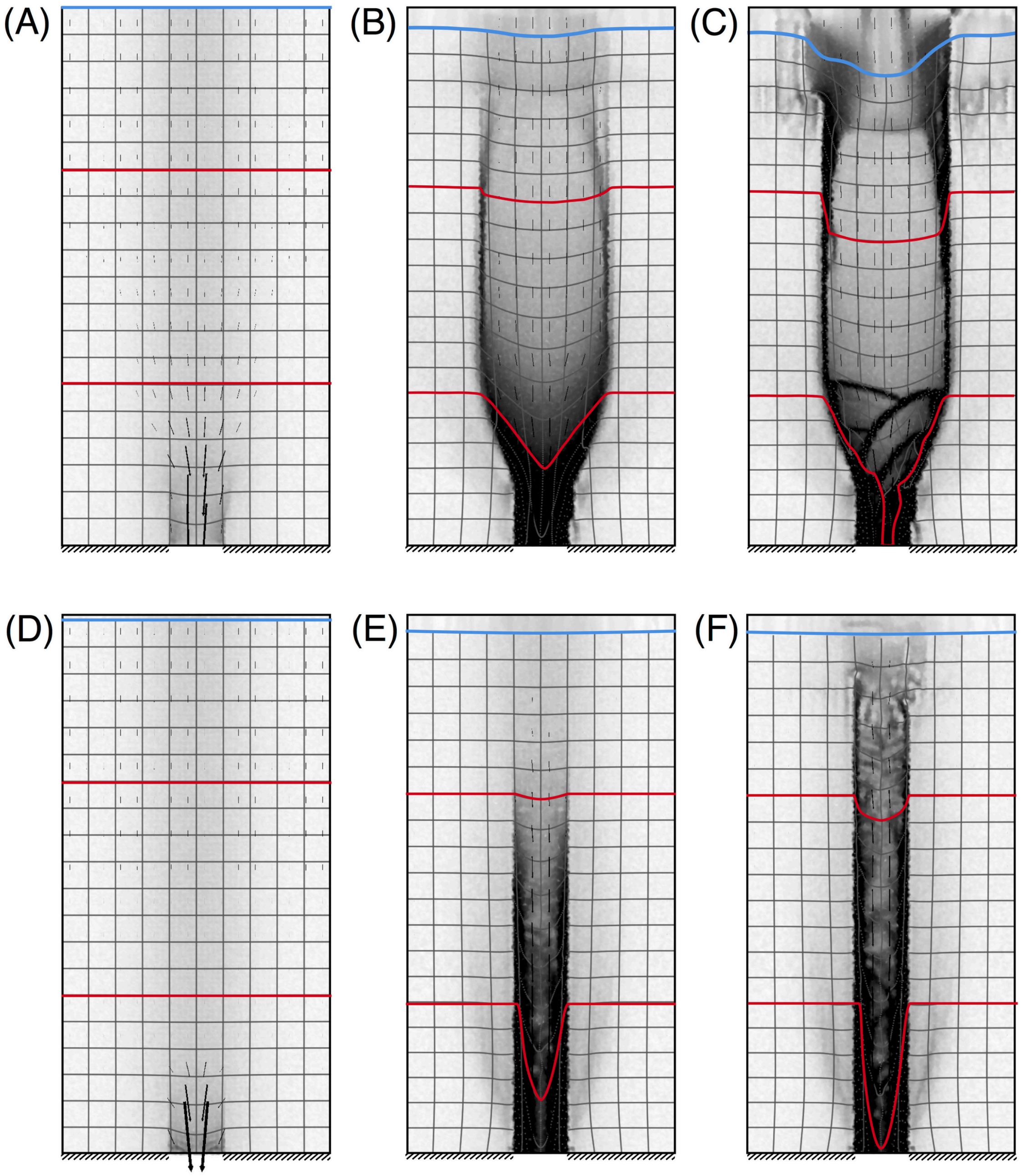
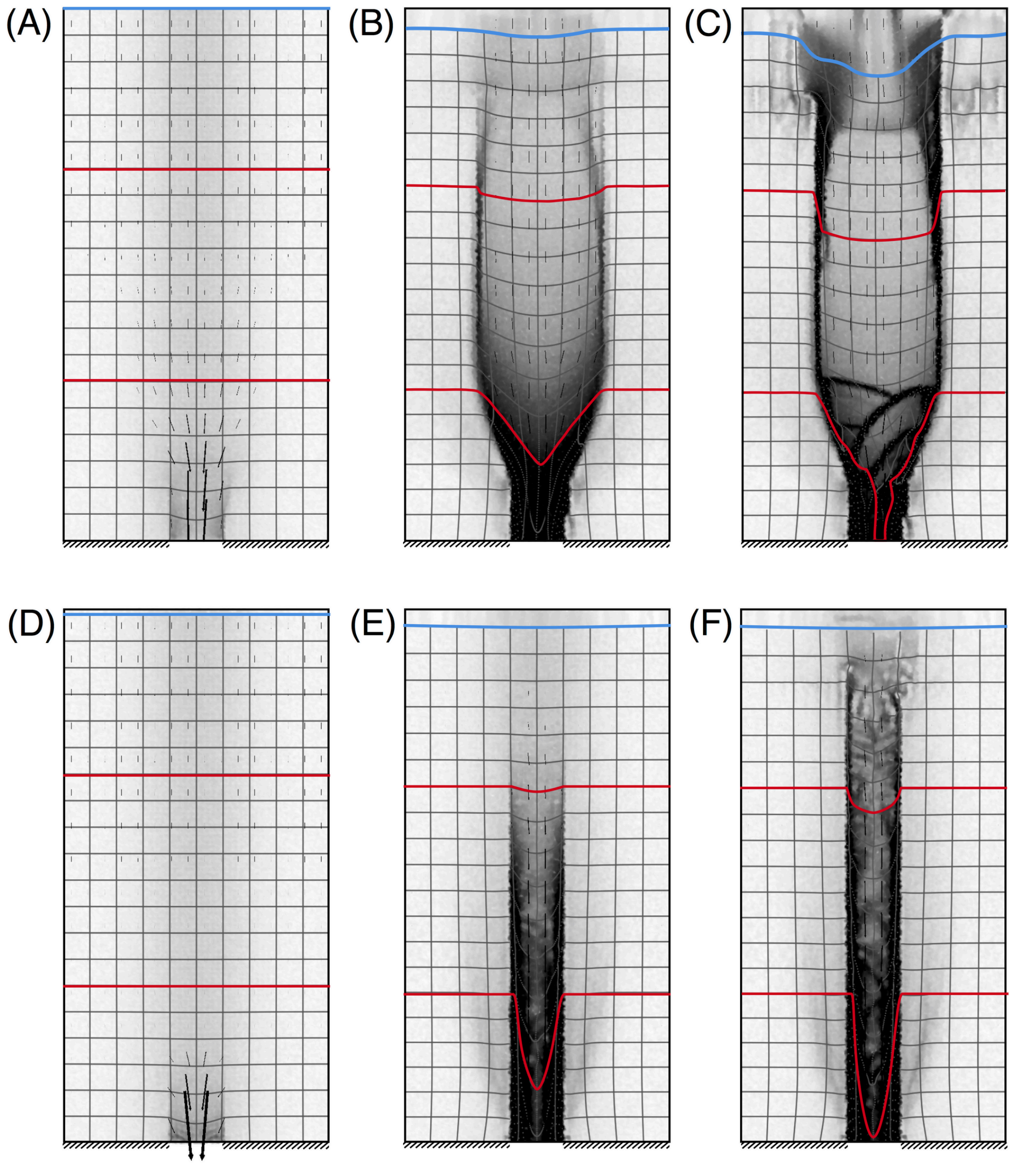


Figure 1: Three snapshots of the total shear strain for a model with low dilatancy (A-C) and a model with high dilatancy (D-F)



1. Update: Published results

A Model for Trap Door Flow from a Deep Container — Computational simulations of trap door flow in cohesive frictional, materials are presented. We focus on flows where the extraction volume is generated by dilatancy. The dilatancy is caused by…

Muhlhaus HB., Moresi L.N. (2019) A Model for Trap Door Flow from a Deep Container. In: Wu W. (eds) Desiderata Geotechnica. Springer Series in Geomechanics and Geoengineering. Springer, Cham, Muhlhaus & Moresi (2019), 2019

References

Muhlhaus, H.-B., & Moresi, L. N. (2019). A Model for Trap Door Flow from a Deep Container. In *Springer Series in Geomechanics and Geoengineering* (pp. 113–118). Springer International Publishing. https://doi.org/10.1007/978-3-030-14987-1_13